Coordinate Plane, Line Plots, and Numerical Patterns — Combined Practice

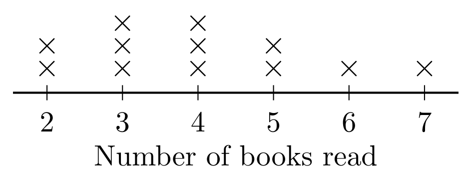
Turning two number patterns into ordered pairs on a grid, summarising a line plot, and running a rule on data you had to read first

Directions:

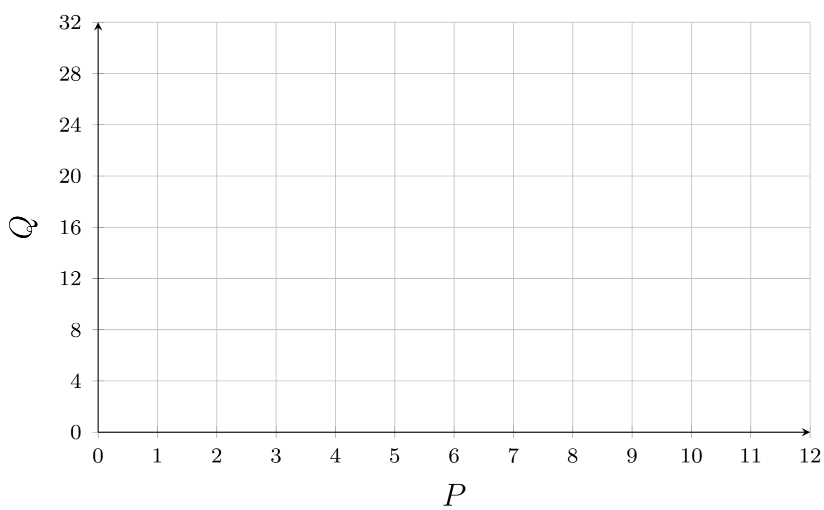
- Every problem needs *two* kinds of information: a pattern rule and a grid, a line plot and a rule, or data and ordered pairs. Decide which two before you start computing.
- **Line Plot A** and **Grid A** are in the box below and are shared by several problems. They carry no problem's numbers: read the numbers you need out of them, and take the rest from the problem you are on.
- Write ordered pairs in the form (x, y) — across first, then up.
- Show your work in the space under each problem, then write the final answer on the answer line.

Line Plot A — Books read last month by Ms. Ortega's class

each × stands for one student



Grid A — for plotting ordered pairs



1. Pattern A adds 2 each time and Pattern B adds 8 each time. Both patterns start at zero. Matching terms are written as ordered pairs (A, B).

Find the B-value of the ordered pair whose A-value is 10.

Answer: ________________

2. Pattern A adds 2 each time and Pattern B adds 8 each time, both starting at zero, exactly as before. Every B-value is the same number of times its A-value.

Find that multiplier, then use it to find the A-value of the ordered pair whose B-value is 72.

Answer: ________________

3. Use **Line Plot A**. How many students read exactly 5 books? Write the count, and also write it as the ordered pair (books read, number of students).

Answer: ________________

4. Use **Line Plot A**. Twelve students are shown on the plot. Find the mean number of books read by a student in this class.

Answer: ____________________

5. Use **Line Plot A**. The reading club gives 5 points for every book a student reads. How many points did the student who read the most books earn?

Answer: ____________________

6. Pattern P starts at one and adds three each time. Pattern Q starts at three and adds nine each time.

Write the first four matching ordered pairs (P, Q), plot them on **Grid A** above, and describe in words how each Q-value is related to its P-value.

Answer: ____________________

7. Use **Line Plot A**. Find the median number of books read.

Then state whether that median is greater than, less than, or equal to the mean you found earlier, and say what that comparison tells you about the shape of the plot.

Answer: ____________________

8. Use **Line Plot A**. Next month every student in the class plans to read 2 more books than they read last month.

What will the new mean number of books be? Explain why you do not have to add up all twelve students' new totals to find it.

Answer: __________________

9. Pattern S starts at zero and adds 5 each time. Pattern T starts at zero and adds 15 each time. Sara says the ordered pair $(20, 45)$ is one of the matching pairs of these two patterns.

Find the correct T-value that goes with the S-value 20, and explain what Sara did wrong.

Answer: __________________

10. Use **Line Plot A**. Find the total number of books read by the whole class last month. Then write the class as one ordered pair: (number of students, total books read).

Answer: ________________

11. The school gives the class one bonus point for every book beyond 40 that the whole class reads in a month. Use the class total you found from **Line Plot A**.

How many bonus points does the class earn?

Answer: ________________

12. Challenge. Suppose the class total grows by six books every month, starting from last month's class total.

Make a table of the class total for the first four months and write each month as an ordered pair (month number, class total). Then describe the pattern in those ordered pairs, and explain what the four plotted points would look like on a grid.

Answer: ________________